

Lab 3: Architecture Selection and Justification

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Architecture Selection:

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We chose the **Layered Architecture** for the Student Club Event Ticketing & Budget Portal System.

Architectural Choice & Reasons: We selected the Layered Architecture pattern because it is "good for traditional applications" and provides a "clear separation of concerns".

1. **Reason 1 (Simplified Development):** This system fits perfectly into standard Presentation (UI), Business (Order/Budget Logic), and Data (Database) layers. This makes the system "easy to understand" and straightforward to develop, as the frontend developers can work on the UI while the backend team handles the core logic and database schemas.
2. **Reason 2 (Centralized Management):** Unlike microservices, which introduce "operational complexity" and "data consistency challenges", a layered approach allows us to easily maintain ACID compliance in our central database, which is critical when handling digital currency ticketing and budget approvals.

Security Advantage:

Layered architecture enforces strict access control between tiers. The Presentation Layer (User Interface) cannot directly communicate with the Data Layer (Database). All requests must pass through the Business Layer (Order Manager / Budget Component), where authentication, input validation, and business rules are enforced, protecting the database from direct exposure to front-end attacks like SQL injection.

Performance Benefit: By avoiding a distributed microservices setup, we eliminate the "network latency" that occurs when multiple independent services have to communicate over a network. In a layered architecture, the Business Layer components can process ticket purchases and query the Data Layer instantly within the same server environment, ensuring fast response times for attendees.